RbsB (PP_2454, Q88K38): Periplasmic D-Ribose-Binding Protein of a High-Affinity ABC Importer in *Pseudomonas putida* KT2440

Summary

RbsB (gene *rbsB*, ordered locus PP_2454, UniProt Q88K38) is the periplasmic, non-catalytic substrate-binding protein (SBP) of a high-affinity D-ribose ATP-binding cassette (ABC) import system in *Pseudomonas putida* KT2440. Its molecular job is to scavenge free D-ribose in the periplasmic space and, through a large ligand-induced "Venus-flytrap" hinge-bending closure, present the captured sugar to the cognate membrane permease (RbsC) so that the ATPase (RbsA) can power its translocation into the cytoplasm. RbsB itself catalyzes no chemical reaction — it is a receptor/adapter whose function is molecular recognition and delivery. This identity rests on three independent lines of evidence: (1) *rbsB* heads a complete, dedicated ribose operon encoding every component needed to import and catabolize D-ribose; (2) the protein is a clear ortholog (39.7% identity) of the structurally and biochemically characterized *E. coli* ribose-binding protein, with near-complete conservation of the ribose-contacting pocket residues; and (3) the operon is controlled by a dedicated ribose-responsive LacI-family repressor (RbsR), the classic regulatory signature of a sugar-specific catabolic import module.

Functionally, RbsB operates **in the periplasm**. It is synthesized with an N-terminal Sec signal peptide (residues 1–29) that is cleaved to yield the mature periplasmic chain (residues 30–319), a single bilobed periplasmic-binding-protein domain. Downstream of binding, imported D-ribose is phosphorylated to ribose-5-phosphate by ribokinase (RbsK, EC 2.7.1.15), feeding directly into the pentose-phosphate pathway. The operon also carries a D-ribose pyranase (RbsD, EC 5.4.99.62) that interconverts the pyranose and furanose ring forms of ribose, and a ribonucleoside hydrolase that can liberate free ribose from nucleosides.

The primary role is therefore **transport-associated substrate capture, not enzymatic catalysis**. In addition to its transport role, homologous ribose-binding proteins in *E. coli* double as the receptor for ribose chemotaxis, feeding ligand-bound protein to the Trg chemoreceptor.

This secondary role is plausible for *P. putida* RbsB by orthology — and KEGG maps PP_2454 to the bacterial chemotaxis pathway — but it has not been experimentally demonstrated in *P. putida* and remains an inference. **Gene-identity verification: PASSED** — the symbol *rbsB*, organism, protein family (bacterial solute-binding protein 2), and domains (Peripla_BP_I / SBP_2 / PF13407) are all mutually consistent, and PP_2454 sits inside a complete, canonical *rbs* operon; there is no evidence of gene-symbol ambiguity.

Key Findings

Finding 1 — *rbsB* is the substrate-binding subunit of a complete, dedicated D-ribose ABC import operon

The genomic context of PP_2454 in *P. putida* KT2440 (KEGG genome `ppu`) reveals a canonical, contiguous *rbs* operon in which every functional element of a ribose import-and-catabolism module is present and adjacent:

Locus	Gene	KEGG KO	Product	Function
PP_2454	**rbsB**	K10439	Ribose ABC transporter substrate-binding protein	Periplasmic D-ribose capture (**target of this report**)
PP_2455	rbsA	K10441	ABC transporter ATP-binding protein (EC 7.5.2.7)	ATP hydrolysis powering transport
PP_2456	rbsC	K10440	Membrane permease	Transmembrane translocation channel
PP_2457	rbsR	K02529	LacI-family transcriptional repressor	Ribose-responsive regulation
PP_2458	rbsK	K00852	Ribokinase (EC 2.7.1.15)	D-ribose → D-ribose-5-phosphate
PP_2459	rbsD	K06726	D-ribose pyranase (EC 5.4.99.62)	Pyranose/furanose interconversion
PP_2460	—	—	Ribonucleoside hydrolase	Ribose release from nucleosides

This is the textbook architecture of a bacterial pentose-import operon. The co-localization of a solute-binding protein, an ATPase, a permease, a dedicated repressor, and the downstream catabolic kinase is diagnostic: RbsB cannot be a stand-alone enzyme, because the operon supplies no active site to it and instead surrounds it with a transport apparatus and a phosphorylating enzyme. UniProt assigns Q88K38 to KEGG orthology **K10439, "ribose transport system substrate-binding protein,"** and to Pfam **PF13407 (Peripla_BP_4 / SBP_2 domain)** with InterPro signatures **IPR028082 (Periplasmic binding protein-like I)** and **IPR025997 (SBP_2)** — exactly the fold family used by periplasmic sugar receptors. The presence of ribokinase and D-ribose pyranase in the operon leaves no reasonable alternative substrate: the module is dedicated to D-ribose acquisition and catabolism. Collectively this establishes RbsB as the periplasmic recognition component of the RbsABC importer.

Finding 2 — RbsB is a periplasmic "Venus-flytrap" receptor that binds D-ribose and delivers it to the permease

RbsB belongs to the periplasmic binding protein (PBP) superfamily, whose members all operate by the same physical principle: two α/β domains connected by a hinge close around the ligand like a Venus flytrap, and downstream partners (the ABC permease and, in some systems, the chemoreceptor) recognize only the closed, ligand-bound conformation.

Several features anchor this interpretation for Q88K38. First, the protein carries a cleaved N-terminal signal peptide (residues 1–29), producing a mature chain (residues 30–319) that resides in the **periplasm / cell envelope** — the correct compartment for a receptor that must intercept sugar after it crosses the outer membrane but before it reaches the inner-membrane transporter. Second, the mature protein consists essentially of a single periplasmic-binding-protein domain (≈ residues 35–299), consistent with the two-lobe SBP architecture.

The mechanistic paradigm comes from the *E. coli* ribose-binding protein (RBP), which is periplasmic and "binds to ribose and mediates transport and chemotaxis" (PMID: 8878033). Structural studies of RBP and close homologs show that binding is coupled to a large open→closed hinge-bending domain motion. Elastic-network modelling of *E. coli* RBP identifies discrete hinge residues (Ser103, Gln235, Asp264) about which the two domains rotate, bend, and twist as rigid bodies, generating the closed ligand-bound state while the internal structure of each domain remains largely intact (PMID: 23698778). The crystal structure of the *Thermotoga maritima* ribose-binding protein — 39% identical to *E. coli* RBP — confirms that polar ligand interactions and the global ligand-induced conformational change are conserved across large evolutionary distances, even when local side-chain rearrangements differ (PMID: 19019243). The same hinge-closure mechanism, with variations in hinge residues, is documented for the related glucose/galactose-binding protein (PMID: 17473016) and the D-allose-binding protein (PMID: 11825912), demonstrating this is the universal operating mode for the receptor class. KEGG maps PP_2454 to both **ppu02010 (ABC transporters)** and **ppu02030 (bacterial chemotaxis)**, mirroring the dual transport/taxis role established for the *E. coli* ortholog.

Finding 3 — RbsB is a clear ortholog of *E. coli* ribose-binding protein with a conserved D-ribose-binding pocket

A Needleman–Wunsch global alignment of Q88K38 (*P. putida*, 319 aa) against P02925 (*E. coli* RbsB, 296 aa) yields **115/290 identical positions** = **39.7% identity** over the aligned length — comfortably within the orthology range and essentially identical to the *E. coli*/*T. maritima* RBP relationship (39%) that preserves full ribose-binding function. Critically, the residues that directly contact ribose in the *E. coli* structure are conserved in the *P. putida* protein:

E. coli RBP residue	*P. putida* RbsB residue	Conserved?
Asn13	Asn44	✔
Phe15	Phe46	✔
Asp89	Asp130	✔
Arg90	Leu131	✘ (substituted)
Arg141	Arg183	✔
Asn190	Asn232	✔
Asp215	Asp258	✔
Gln235	Gln278	✔

Seven of eight ribose-contacting residues are conserved (the single exception being Arg90→Leu131). This degree of pocket conservation strongly supports **D-ribose as the physiological substrate** of *P. putida* RbsB — direct structural-bioinformatic evidence independent of the genomic-context argument. Both proteins share the SBP_2 (Peripla_BP) fold and carry an N-terminal Sec signal peptide directing periplasmic export. Because these pocket residues form the hydrogen-bonding network that reads out the specific stereochemistry of the ribose sugar, their conservation across a large evolutionary distance is far stronger evidence of substrate identity than overall sequence similarity alone. An AlphaFold model (AlphaFoldDB Q88K38) is expected to reproduce the bilobed PBP fold.

Finding 4 — The operon is controlled by a dedicated ribose-responsive LacI-family repressor (RbsR)

Within the *rbs* operon, PP_2457 (*rbsR*) encodes a LacI-family transcriptional repressor (KEGG K02529). In characterized homologs, RbsR represses the ribose gene cluster and its DNA-binding activity is relieved by a ribose-derived effector, providing tight, substrate-inducible control of RbsB expression:

- In *Bifidobacterium breve* UCC2003, "the promoter upstream of rbsABCDK is negatively controlled by RbsR(His) binding to an 18 bp inverted repeat and that RbsR(His) binding activity is modulated by D-ribose" (PMID: 21255330). The same study confirmed that the operon's ribokinase specifically phosphorylates D-ribose to ribose-5-phosphate (EC 2.7.1.15), cementing the pathway logic.
- In *Corynebacterium glutamicum* ATCC 13032, RbsR represses the *rbsRACBD* operon; deletion of *rbsR* increases operon mRNA, and the physiological effector is likely **ribose-5-phosphate** (or a derivative), since a ribokinase double mutant shows no derepression by ribose (PMID: 19118356). Deletion of the structural genes *rbsACBD* abolished ribose uptake, directly linking this gene set to ribose transport.

The presence of a dedicated, co-transcribed RbsR repressor is a hallmark of a substrate-specific sugar-import module: RbsB is produced on demand when ribose (or its phosphorylated derivative) is available, as expected for a high-affinity scavenging receptor rather than a constitutively expressed housekeeping protein. This mirrors the broader *P. putida* KT2440 strategy of dedicated LacI/GntR-type transcription factors controlling clustered catabolic genes (PMID: 39589324).

Mechanistic Model / Interpretation

The four findings converge on a single, coherent mechanistic picture in which RbsB is the entry-point receptor of a linear import-then-catabolize pathway:

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  PERIPLASM                             INNER MEMBRANE        CYTOPLASM

┌───────────────┐
│  D-ribose     │
│  (free sugar) │
└──────┬────────┘
       │  capture (Venus-flytrap closure)
       ▼
┌───────────────┐           open→closed
│   RbsB        │   ──────hinge motion───────►  delivers closed,
│  (PP_2454)    │                                ligand-bound form
│  periplasmic  │                                       │
│  SBP          │                                       ▼
└───────────────┘                           ┌────────────────────────┐
                                            │  RbsC permease (PP_2456)│
                                            │  channel in membrane    │
                                            └───────────┬────────────┘
                                                        │  ATP-driven translocation
                                            ┌───────────▼────────────┐
                                            │  RbsA ATPase (PP_2455)  │
                                            │  EC 7.5.2.7             │
                                            └───────────┬────────────┘
                                                        ▼
                                          D-ribose in cytoplasm
                                                        │
                              RbsD pyranase (PP_2459)   │  ring interconversion
                              EC 5.4.99.62              ▼  (pyranose ⇌ furanose)
                                          RbsK ribokinase (PP_2458)
                                                EC 2.7.1.15
                                                        │
                                                        ▼
                                            D-ribose-5-phosphate
                                                        │
                                                        ▼
                                          PENTOSE-PHOSPHATE PATHWAY
                                          (nucleotide biosynthesis,
                                           NADPH, central metabolism)

 Regulation:  RbsR (PP_2457, LacI-family) represses the whole operon;
              ribose / ribose-5-phosphate relieves repression (induction).
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Step-by-step interpretation:

1. **Recognition (periplasm).** Free D-ribose that has crossed the outer membrane is captured by RbsB. The two lobes of the SBP fold close around the sugar via a hinge-bending motion, forming a high-affinity, stereospecific complex. The conserved pocket residues (Asn44, Phe46, Asp130, Arg183, Asn232, Asp258, Gln278) hydrogen-bond and stack against the ribose ring, enforcing selectivity for ribose over other sugars.
2. **Delivery (inner membrane).** Only the closed, ligand-loaded conformation of RbsB is recognized by the RbsC permease. This conformational proofreading ensures the transporter fires only when cargo is present.
3. **Translocation (across the inner membrane).** RbsC provides the transmembrane pathway; RbsA hydrolyzes ATP (EC 7.5.2.7) to power import of D-ribose into the cytoplasm.
4. **Catabolic commitment (cytoplasm).** RbsD (D-ribose pyranase, EC 5.4.99.62) equilibrates the ring forms of ribose to supply the substrate configuration preferred by the kinase. RbsK (ribokinase, EC 2.7.1.15) then phosphorylates D-ribose to D-ribose-5-phosphate, which is simultaneously a metabolic trap (preventing efflux) and the entry metabolite for the pentose-phosphate pathway, nucleotide biosynthesis, and NADPH generation. In *P. putida*, which relies on the Entner–Doudoroff pathway rather than glycolysis for hexose catabolism, ribose-5-phosphate provides biosynthetic precursors and, through PPP interconversions, carbon/energy.
5. **Regulation (transcription).** The LacI-family repressor RbsR keeps the operon off in the absence of ribose. When ribose (or ribose-5-phosphate) accumulates, it binds RbsR, relieving repression and inducing synthesis of the whole module — including RbsB.

Within this model, **RbsB's precise role is molecular recognition and hand-off**: it is the affinity-determining, specificity-conferring receptor that converts the presence of extracellular ribose into a productive transport event. It performs no chemistry itself; its "output" is a conformational signal (the closed state) that licenses transport. The location of its function is unambiguously the **periplasm/cell envelope**. Its pathway membership is **carbohydrate (pentose) uptake and catabolism feeding the pentose-phosphate pathway**, with a plausible but unconfirmed secondary role in **ribose chemotaxis**.

Evidence Base

PMID	Title (abbreviated)	How it supports the findings
8878033	*Genetically probing the regions of ribose-binding protein involved in permease interaction*	Establishes the *E. coli* RBP paradigm: periplasmic, "binds to ribose and mediates transport and chemotaxis"; identifies distinct surface regions for permease vs. chemoreceptor interaction. Anchors the dual transport/taxis role and receptor-adapter function (F002).
23698778	*Analysis of conformational motions ... for E. coli ribose-binding protein (elastic network models)*	Defines the open→closed hinge-bending mechanism and hinge residues (Ser103, Gln235, Asp264); domains move as rigid bodies. Mechanistic basis for the Venus-flytrap model (F002).
19019243	*Ligand-induced conformational changes in a thermophilic ribose-binding protein*	*T. maritima* RBP is 39% identical to *E. coli* RBP yet conserves polar ligand interactions and global conformational change — validating conservation of ribose binding at ~40% identity, precisely the *P. putida*/ *E. coli* relationship (F002, F003).
17473016	*Conformational changes of glucose/galactose-binding protein ...*	Independent PBP-superfamily example confirming hinge-bending closure mediates ligand binding, transport, and signaling; hinge residues vary between family members (F002).
11825912	*Hinge-bending motion of D-allose-binding protein from E. coli*	Closely related receptor; open and closed structures show torsional hinge changes with intact domains, and explicitly contrasts with ribose-binding protein's pattern — reinforces the family mechanism (F002).
19118356	*Characterization of the LacI-type repressor RbsR ... in C. glutamicum*	Deletion of *rbsACBD* abolishes ribose uptake (transport function); RbsR represses the *rbs* operon and is derepressed via ribose-5-phosphate. Supports F001 (transport operon) and F004 (RbsR regulation, effector identity).
21255330	*Ribose utilization by ... Bifidobacterium breve UCC2003*	*rbsACBDK* cluster essential for ribose utilization; RbsR binds an 18-bp inverted repeat and is modulated by D-ribose; RbsK is a D-ribose-specific ribokinase (EC 2.7.1.15). Supports F001, F004, and the downstream phosphorylation step.

PMID	Title (abbreviated)	How it supports the findings
39589324	*GnuR represses glucose and gluconate catabolism in P. putida KT2440*	*P. putida* KT2440 regulatory context: clustered catabolic genes controlled by dedicated LacI/GntR-type TFs, and Entner–Doudoroff-centric metabolism — consistent with the RbsR-controlled *rbs* cluster architecture (F004, background).
36354357	*Glucose-6-phosphate dehydrogenase ZwfA in Pseudomonas*	Background on *Pseudomonas* central carbon metabolism (Entner–Doudoroff and oxidative pentose-phosphate pathways), the downstream destination of imported ribose-derived ribose-5-phosphate (mechanistic context).

Overall weight of evidence. The identity of RbsB as a periplasmic D-ribose-binding transport protein is supported by convergent structural, evolutionary, genomic, and regulatory evidence. The *direct* experimental characterization comes from orthologs (*E. coli*, *T. maritima*, *B. breve*, *C. glutamicum*), while the *P. putida*-specific evidence is bioinformatic/comparative (operon structure, sequence orthology, pocket conservation, domain assignment). No evidence contradicts the ribose-binding assignment.

Limitations and Knowledge Gaps

1. **No direct biochemistry on *P. putida* Q88K38.** There is no published purification, binding assay (e.g., ITC/fluorescence Kd for D-ribose), or crystal structure of the *P. putida* protein specifically. Substrate identity is inferred from orthology and pocket conservation, not measured directly in this organism.
2. **One binding-pocket residue is not conserved.** The *E. coli* Arg90 is replaced by Leu131 in *P. putida* RbsB. In *E. coli*, Arg90 participates in ribose coordination; its substitution could subtly alter affinity or specificity. Whether *P. putida* RbsB retains identical ribose affinity, or has shifted specificity (e.g., toward a ribose analog or related pentose), is untested.
3. **The chemotaxis role is inferred, not demonstrated.** KEGG maps PP_2454 to the chemotaxis pathway and the *E. coli* ortholog serves the Trg chemoreceptor, but *P. putida* KT2440 has a different chemoreceptor repertoire, and no ribose-taxis phenotype linked to RbsB has been reported. Treat the secondary signaling role as a hypothesis.

4. **RbsR effector and operator not mapped in *P. putida*.** The physiological inducer (D-ribose vs. ribose-5-phosphate) and the operator sequence for PP_2457 RbsR are inferred from *C. glutamicum*/*B. breve* homologs, not experimentally defined in KT2440.
5. **Physiological relevance of ribose to *P. putida* growth is unquantified.** It is unknown how strongly KT2440 relies on ribose as a carbon source, whether the *rbs* operon is induced under laboratory or environmental conditions, or whether the ribonucleoside hydrolase (PP_2460) supplies ribose from nucleoside salvage as the primary in vivo substrate source.
6. **Signal-peptide and localization boundaries are predicted.** The signal peptide (1–29) and mature-chain boundaries derive from UniProt sequence-based prediction; periplasmic localization, while highly likely, has not been experimentally confirmed for this protein.

Proposed Follow-up Experiments / Actions

1. **Direct binding measurement.** Overexpress and purify mature RbsB (residues 30–319, His-tagged) and measure D-ribose affinity by isothermal titration calorimetry or intrinsic/extrinsic fluorescence. Include a panel of related sugars (D-arabinose, D-xylose, D-ribose-5-phosphate, D-allose, D-glucose) to define specificity and test whether the Arg90→Leu131 substitution has shifted selectivity.
2. **Structure determination.** Solve the crystal or cryo-EM structure of RbsB in apo and ribose-bound states to confirm the two-lobe SBP fold, verify the hinge-bending closure, and directly visualize the ribose-contact residues (Asn44, Phe46, Asp130, Arg183, Asn232, Asp258, Gln278). AlphaFold modelling plus docking would be a rapid first pass.
3. **Genetic transport phenotype.** Construct a clean *rbsB* (PP_2454) deletion in KT2440 and assay growth on D-ribose as sole carbon source and labeled-ribose uptake rates, with complementation. Compare against Δ*rbsA*/Δ*rbsC* to confirm the SBP is essential for high-affinity uptake.
4. **Regulation.** Map the RbsR (PP_2457) operator by EMSA/DNase footprinting on the *rbsB* promoter, determine the effector (ribose vs. ribose-5-phosphate) using ribokinase-mutant backgrounds, and quantify operon induction by RT-qPCR/RNA-seq during growth on ribose vs. glucose.

5. **Chemotaxis test.** Perform capillary or plate chemotaxis assays with D-ribose gradients in wild-type vs. Δ*rbsB* KT2440 to determine whether RbsB contributes to ribose taxis, and identify the candidate chemoreceptor partner.
6. **Substrate source in vivo.** Test whether the co-operonic ribonucleoside hydrolase (PP_2460) supplies ribose from nucleosides by comparing growth/uptake on ribose vs. ribonucleosides in relevant mutants, clarifying the ecological/physiological role of the module.

Conclusion. All available evidence — operon architecture, ortholog-level sequence identity with conserved binding-pocket residues, SBP_2/Peripla_BP domain assignment, periplasmic localization signal, and dedicated RbsR regulation — identifies RbsB (PP_2454, Q88K38) as the periplasmic D-ribose-binding receptor of the RbsABC high-affinity ABC importer in *Pseudomonas putida* KT2440. It is a non-catalytic substrate-recognition/adapter protein that captures D-ribose in the periplasm and delivers it to the membrane permease for ATP-driven import into a ribokinase-terminated catabolic pathway feeding the pentose-phosphate pathway, with a plausible but experimentally unconfirmed secondary role in ribose chemotaxis.
